quiz

Please take out a piece of paper, write your name and student ID, and complete the following questions. Submit your work after 15 minutes.

Suppose an e-commerce platform has two users A and B with different product browsing histories. The system records their viewed product IDs as follows:

- **User A**: {1, 3, 5, 7, 9, 11, 13, 15, 17, 19}
- **User B**: {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}

1) Calculate the **Jaccard Similarity** between the two users.
2) Use **MinHash** to estimate their Jaccard similarity by applying **3 independent random permutations** (*the order of elements is messed up (打乱) 3 times*).

2026 04.28 → 06.19

数据科学与大数据技术的数学基础

Mathematical Foundations of Data Science

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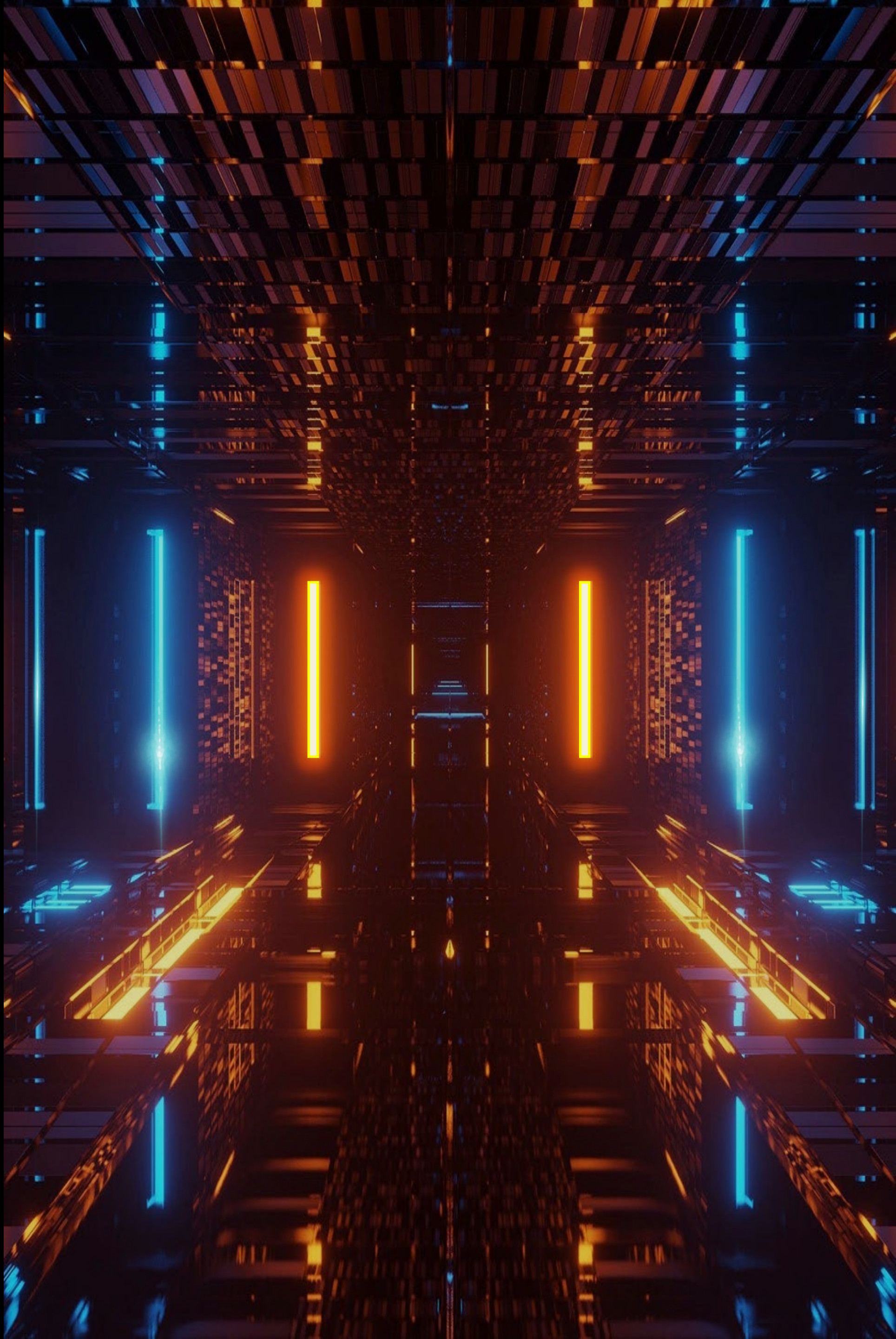


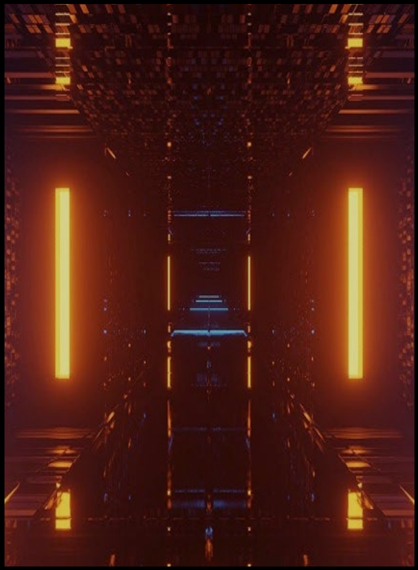
数据科学与大数据技术的数学基础

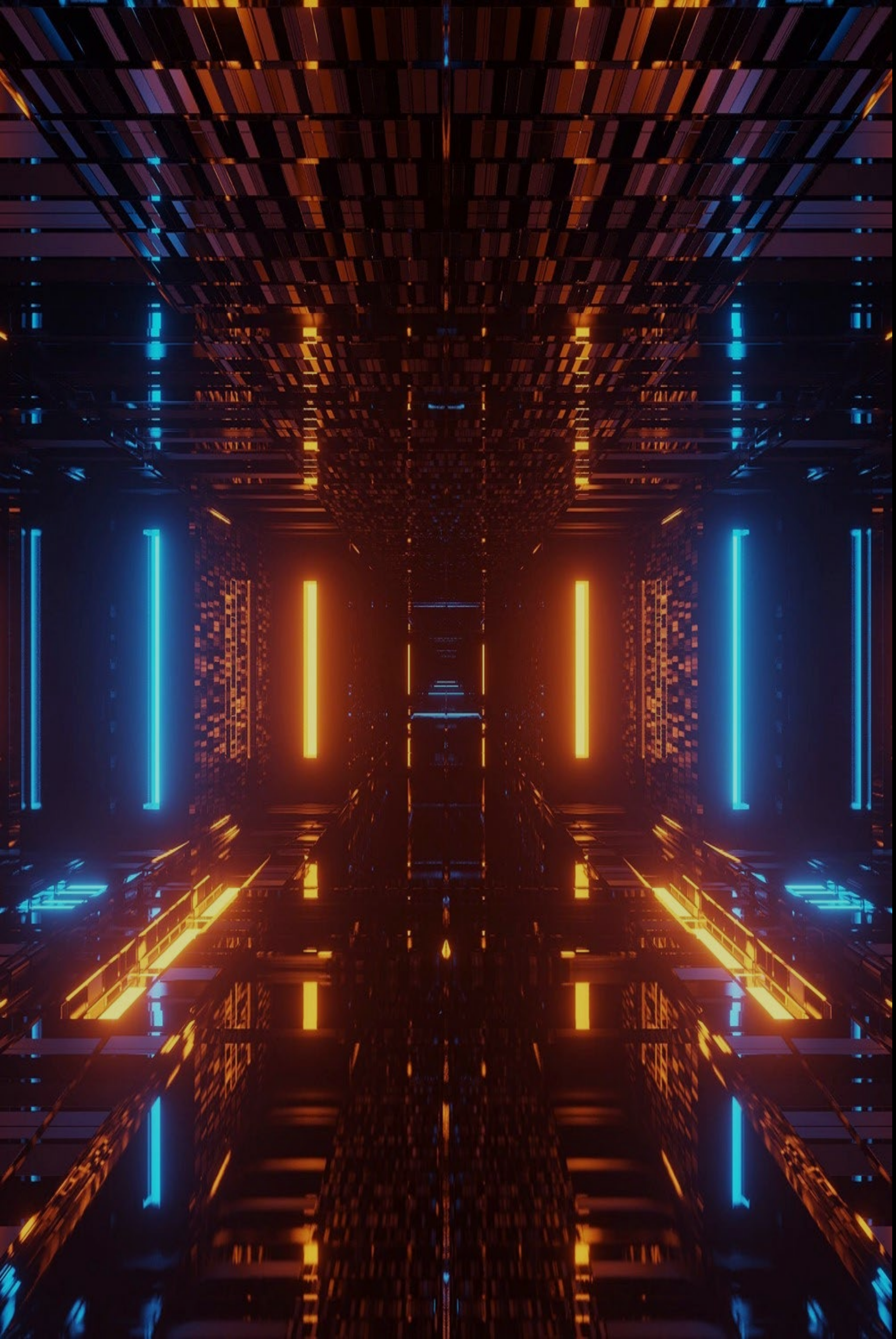
01

Mathematical Foundations of Data Science

/03 泛化与正则

Generalization and Regularization





Contents

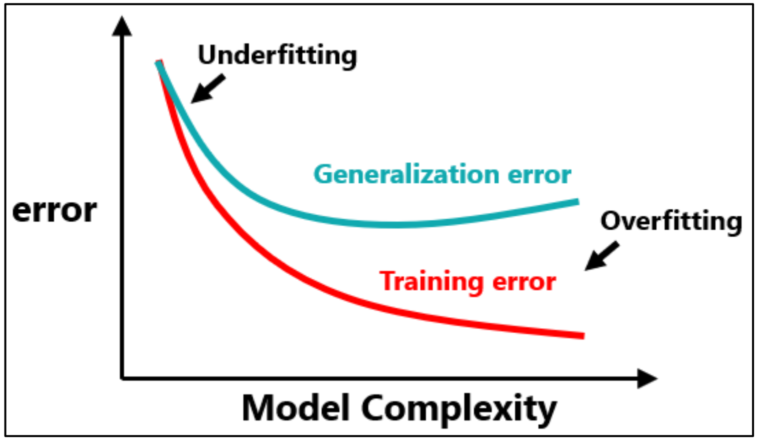
3.1 What is generalization

Generalization

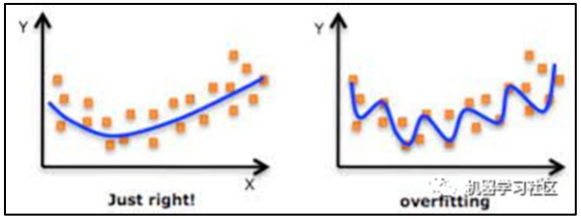
This mastery of the rules is the generalization ability.

My roommate is good at summarizing the rules of doing questions, she has good generalization ability;

while I only know how to brush the questions but do not master the rules, so I have poor generalization ability.



The fundamental problem of **machine learning (deep learning)** is the opposition between **optimization** and **generalization**.



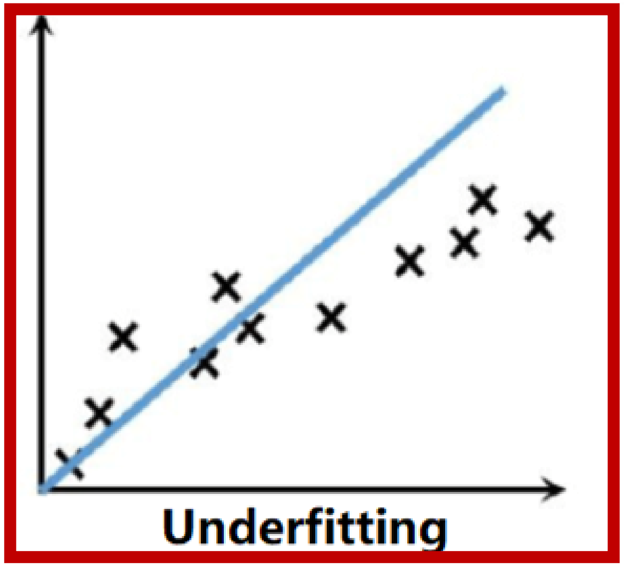
Optimization refers to **adjustment** the model to get the best performance on the **training data** (learning in machine learning)

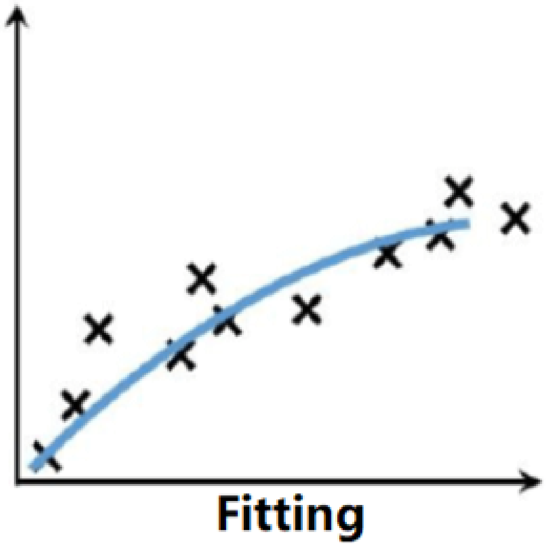
Generalization, on the other hand, refers to how well a **trained** model performs on **data that has never seen before**.

Ideal and Reality

The goal of machine learning is **get good generalization**, but we can no control over the generalization and only **adjust the model based on the training data.**

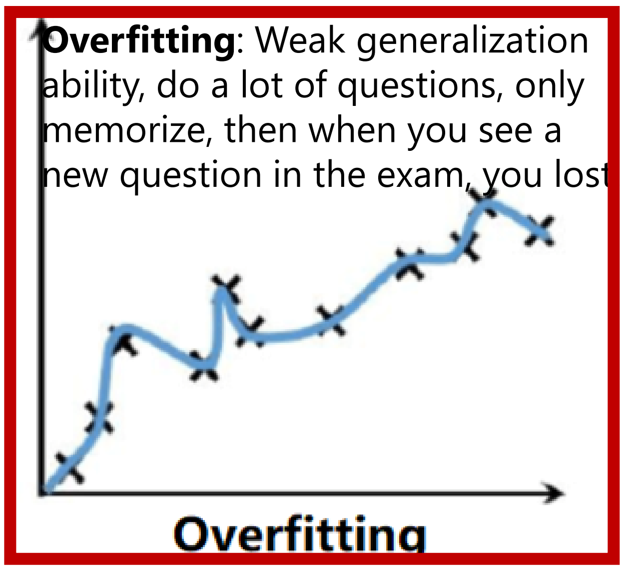
What is generalization



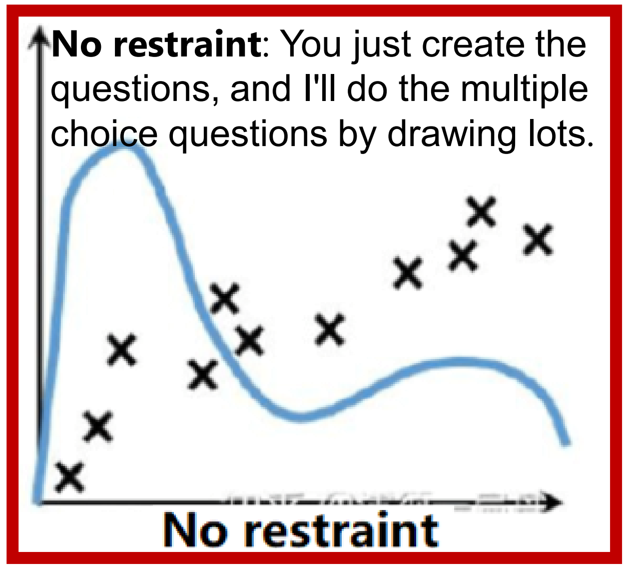


The **data labeled in some of the graphs are marked with “+” plus signs and "-" reduce signs**, and the **predicted classifiers are indicated by straight lines or curves**, which allow you to compare the relationship between the data labels and the position of the lines and think about underfitting, overfitting, fitting, and non-convergence

Overfitting: Weak generalization ability, do a lot of questions, only memorize, then when you see a new question in the exam, you lost



No restraint: You just create the questions, and I'll do the multiple choice questions by drawing lots.



Underfitting: The generalization ability is weak, I have done many questions, but I can't grasp the law, and I can't answer questions no matter old or new.

3.1 What is generalization

For **classification learning algorithms**, we generally divide the sample set into a **training set** and a **test set**:
Training set: learning or training the algorithm model
Test set: evaluating the predictive performance of the trained model on the data

The algorithm first trained on the **training set** → Obtain a model → Evaluate performance on the **test set**

Taking the classification task (give a picture to see if it's a cat) as an example, let's understand some basic concepts related to models and machine learning

1) **Error rate**: The ratio of the number of misclassified samples to the total number of samples. The error rate E=a/m if there are a samples out of m samples that are incorrectly classified.
2) **Accuracy** = 1 - error rate = 1 - a/m.
3) **Error:** the difference between the **predicted output of the learner** and the **true output of the sample**
4) **Training error or empirical error:** the error of the learner on the training set (the ratio of missplit samples of the model **on the training set**)
5) **Test error:** the rate at which the model scores wrongly on the test set
6) **Generalization error:** the expected error of the model on the unknown sample, is the ratio of missplit samples in the unseen data.
7) Therefore, when the sample set is divided, if the data in the training set and the test set do not intersect, **the test error is basically equivalent to the generalization error**

The ideal situation is to **evaluate the generalization error of the candidate models and then select the one with the lowest generalization error**. However, we generally do not get the generalization error directly, and the training error is not suitable as a criterion due to the existence of overfitting phenomenon, so how to perform model evaluation and selection in reality?

The selection of the test set: the test set should be sampled independently from the true distribution of the samples, and the test set should be as mutually exclusive as possible with the training set, i.e., the test samples should not appear in the training set and have not been used in the training process.

3.2 How to understand generalization

1. Generalization capability most directly defined as the difference between training data and real data, where the purpose of training the model is for the model to be tested on completely unfamiliar real data.

Therefore, when performing **cross-validation**, it is important to ensure that the test set and the training set have the **same data distribution**. When the test set and training set have different distributions, **Adversarial Validation**, which mixes the training and test sets, can be used to cope with this.

举个例子，假设你的团队开发了一套能在开发集上运行性能良好，却在测试集上效果不佳的系统。如果此时开发集和测试集的分布相同，那么你就能清楚地明白问题所在：算法在开发集上过拟合了（overfit）。解决方案显然就是去获取更多的开发集数据。

但是如果开发集和测试集服从不同的分布，解决方案就不那么明确了。此时可能存在以下一种或者多种情况：

1. 算法在开发集上过拟合了。
2. 测试集比开发集更难进行预测，尽管算法做得足够好了，却很难有进一步的提升空间。
3. 测试集不一定更难预测，但它与开发集性质并不相同（分布不同）。因此在开发集上表现良好的算法不一定在测试集上也能够取得出色表现。如果是这种情况，大量针对开发集性能的改进工作将会是徒劳的。

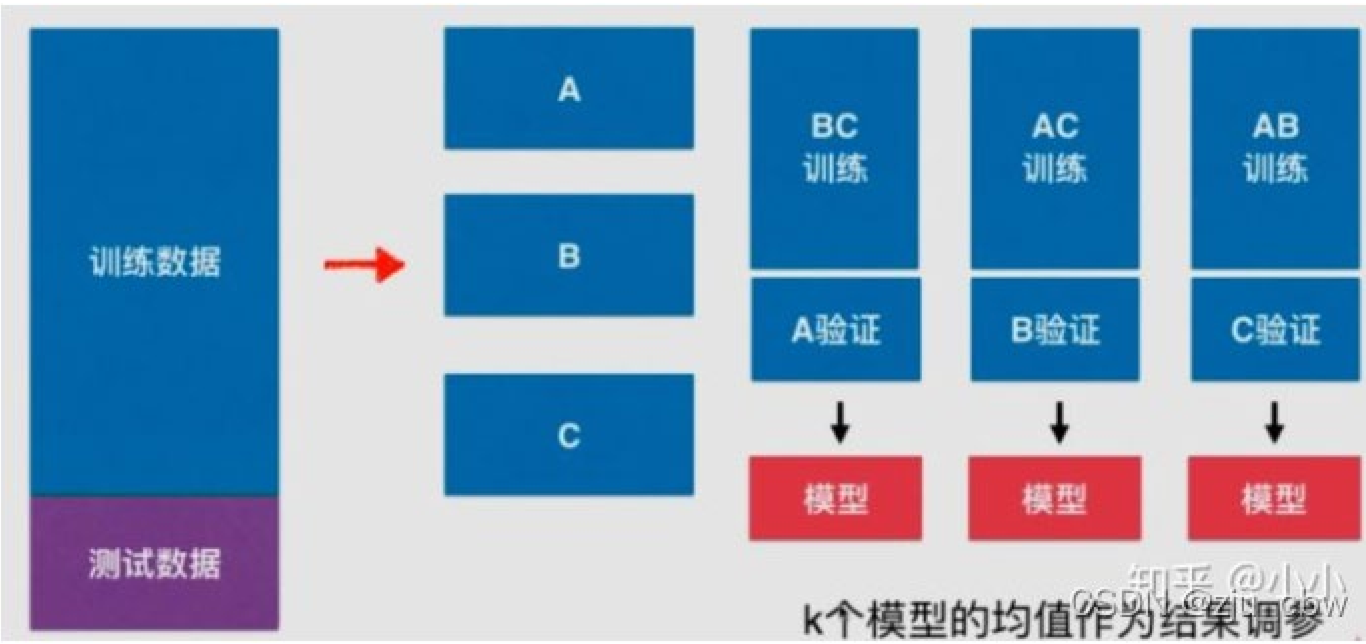
2. The generalization ability can also be seen as the sparsity of the model. As Occam's Razor points out, choosing simple, falsifiable representations makes complex problems simple. In machine learning, a model with generalization capability should have many parameters that are close to 0. In deep learning, it is the matrix to be optimized that should have a preference for sparsity.

? Why smaller parameters indicate simpler models in machine learning

Because the more complex the model, the more it tries to fit all samples, even including some unusual sample points, which can easily cause large fluctuations in the predicted values in smaller intervals, and such large fluctuations also reflect the large derivatives in this interval, which can only be produced by larger parameter values. Thus complex models with relatively large parameter values

PS From cross-validation to Error(误差）、Bias(偏差)、Variance(方差)

On the previous page, we mentioned dividing the data set into training data and test data according to 7:3, ensure that the test set and training set **have the same data distribution**. To improve data availability and reduce overfitting and underfitting problems, **cross-validation** can be used to improve the model generalization ability.



For K, there are two strange descriptions. When **K is large**, we have **less Bias** and **more Variance**, and when **K is small**, we have **more Bias** and **less Variance**.

The method illustrated above is called K-fold Cross Validation: the initial sample is split into K subsamples, a single subsample is kept as data for validating the model, and the other $K-1$ samples are used for training. The cross-validation is repeated K times, once for each subsample, and the results are averaged over K times or by using other combinations, resulting in a single estimation.

PS From cross-validation to Error(误差) 、 Bias(偏差)、Variance(方差)

Error: reflects the accuracy of the overall model

Error = Bias + Variance + Noise

Bias error: reflects the error between the **output of the model on the sample** and the **true value**, i.e., the **accuracy** of the model itself.

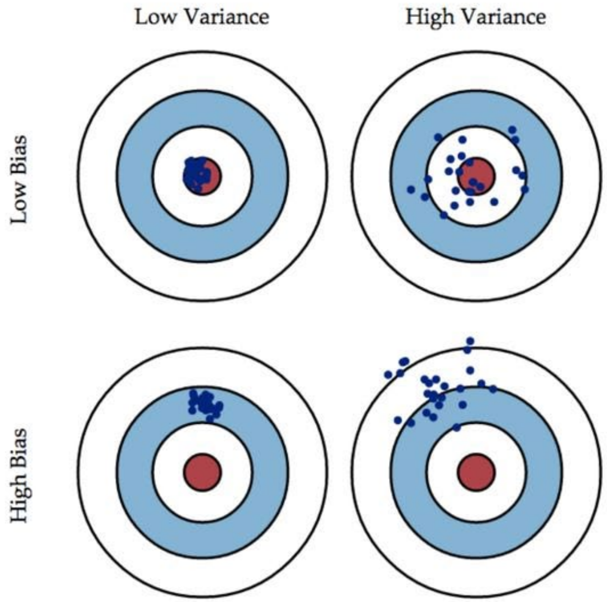
高偏差High Bias：准头差，对眼瞄准

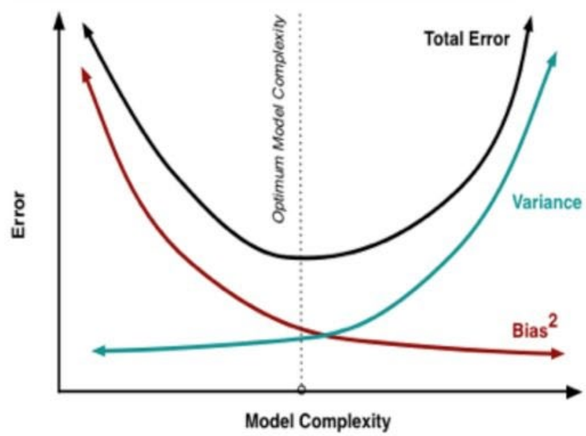
Variance error: reflects the error between each **output and the expected output** of the model, i.e. the **stability** of the model.

高方差High Variance：结果分散，霰弹枪

Bias and Variance are often incompatible. If the Bias of the model is to be reduced, the Variance of the model will be increased to some extent, and vice versa.
The root cause of this phenomenon is that we always want to try to estimate infinite real data with a finite training sample.

准与确





PS Kullback-Leibler Divergence

Kullback-Leibler Divergence (KL Divergence) is a method for measuring the difference between two probability distributions, commonly used in machine learning, information theory, and statistics. It quantifies the **loss of information** **(or approximation error)** when using a distribution *Q* to approximate the true distribution *P*.

Core Definition

For two discrete probability distributions *P* and *Q*, the KL divergence is calculated as:

$$D_{KL}(P||Q) = \sum_i P(i) \log \frac{P(i)}{Q(i)}$$

For continuous distributions:

$$D_{KL}(P||Q) = \int_{-\infty}^{\infty} P(i) \log \frac{P(i)}{Q(i)} di$$

Key Properties:

1.Asymmetry: $D_{KL}(P||Q) \neq D_{KL}(Q||P)$.

1. The former is called **forward KL**, which tends to cover all modes of *P* (commonly used in generative models).

2. The latter is called **reverse KL**, which avoids low-probability regions in Q (often used in approximate inference).

2. **Non-negativity**: $D_{KL}(P||Q) \geq 0$, with equality holding if and only if $P = Q$.

3. **Not a metric**: It violates the triangle inequality and thus is not a true "distance" measure.

Example 1: Coin Toss Experiment (Discrete Distribution)

•**True distribution $\boldsymbol{P}$**: A fair coin with P(heads)=0.5 and P(tails)=0.5.

•**Approximate distribution $\boldsymbol{Q}$**:

A biased coin with P(heads)=0.8 and P(tails)=0.2.

Calculating KL divergence:

$$D_{KL}(P||Q) = 0.5 \log\frac{0.5}{0.8} + 0.5 \log\frac{0.5}{0.2} \approx 0.5 \times (-0.47) + 0.5 \times 0.92 \approx 0.23 \ \ (\text{nats})$$

Interpretation: When approximating the fair coin P with the biased coin Q, an average of **0.23 nats** of additional information is required per event to correct the error.

3.2 How to understand generalization

3. Generate high-fidelity capabilities in the model. Models with generalization capabilities should have the ability to reconstruct features at each level of abstraction.

For example, a CNN can ignore the location of the feature of interest, while a capsule network can ignore whether the feature is rotated or not. By removing more and more irrelevant features, we can ensure the model's ability to accurately generate the features it really cares about. This goes hand in hand with the third point above.

4. the model can effectively ignore trivial features, or find the same features under unrelated variations.

Machine learning often appears an important module, **Dropout**, Dropout is commonly used in the case of large amounts of data, it can be more effective in reducing the occurrence of overfitting, with the effect of integrated learning, to a certain extent, to achieve the effect of regularization.

Dropout

we let the activation value of a certain neuron stop working with a certain probability p (random deactivation) during forward propagation, which makes the model more generalizable because it does not depend too much on certain local features.

First, the random deactivation of the model is ultimately equivalent to getting a lot of different model, but the number of parameters to be trained is constant at this point, with some implication of integrated learning

Second, because of the dropout procedure, two neurons do not necessarily appear in a dropout network every time. The update of the weights no longer depends on the joint action of implicit nodes with fixed relationships, preventing the situation that some features only work with other specific features.

3.2 How to understand generalization

5. The generalization capability can also be seen as the information compression capability of the model.

This explains **why deep learning works**, **information bottleneck**, **the better a model is at compressing features** (dimensionality reduction) , the more likely it is to **make accurate classifications**.

The information compression ability can be **summarized in the 4 explanations of generalization ability above.**

Sparse models finish information compression due to their structure, models with **high generation ability** and **low generalization error** are possible due to information compression, **ignoring irrelevant features** is byproduct of information compression.

2. The generalization ability can also be seen as the sparsity of the model. As Occam's Razor points out, choosing simple, falsifiable representations makes complex problems simple. In machine learning, a model with generalization capability should have many parameters that are close to 0. In deep learning, it is the matrix to be optimized that should have a preference for sparsity.

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6. A final perspective for understanding generalization capabilities is risk minimization.

This is from a game-theoretic perspective, where a model with high generalization ability minimizes its own risk of encountering surprises in a real environment, and therefore generates an internal warning mechanism for unknown features and prepares a response plan in advance.

Empirical risk minimization, ERM

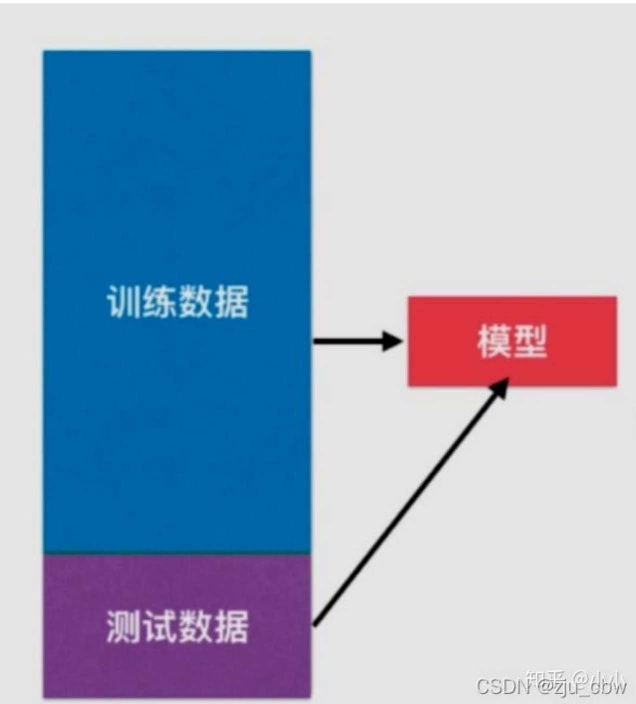
Empirical risk minimization is based on the cost function to minimize the overall cost (loss) function of the training set

Structural risk minimization, SRM

Structural risk minimization (SRM) is a **strategy proposed to prevent overfitting**. **Structural risk minimization is equivalent to regularization**. Structural risk adds a regularization term representing the complexity of the model to the empirical risk to reduce the complexity of the model and improve the predictability of the model.

3.3 Dataset classification

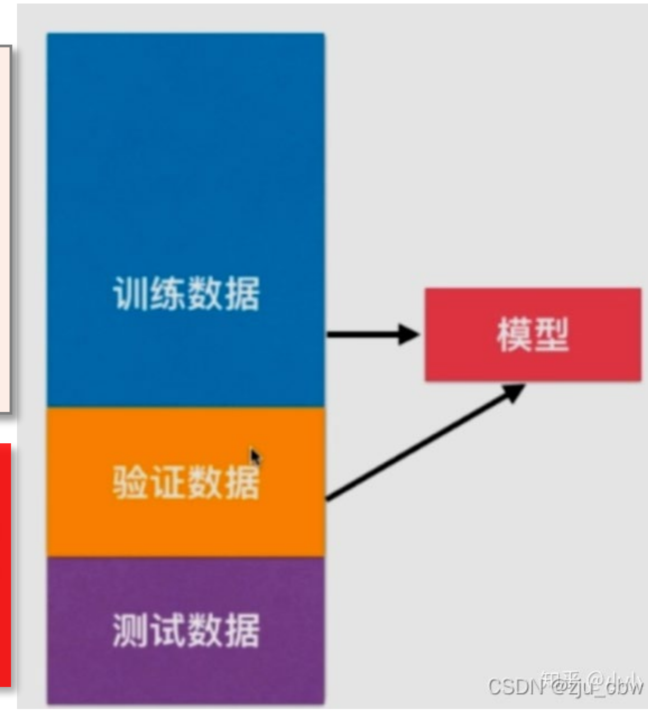
- ✓ the questions we brush before the exam are the **training set**;
- ✓ the questions in the exam are the **test samples** we encounter after modeling;
- ✓ the **generalization** is the **performance of our model encountering the new samples** (which corresponds to the score of the exam)



?

If the model is tuned to the test set, then the model is likely to be overfitted on the test set, or if this is done, it won't reflect the generalization ability of the model.

Validation set also randomness, and it is likely that overfitting will occur because of this one validation set, so cross-validation is generated again.



1) The general practice is to **divide the dataset into a training set and a test set**, train the model with training set, and test generalization ability of model with test set. We also **adjust the model according to the accuracy of the test set**.

2) So it is more practice to **divide the dataset into training set, validation set, and test set.** The training set is used to **train** the model, the validation set to **adjust** the model, and the test set to **test** the generalization ability of the model.

3.3 Dataset classification

The dataset can be divided into:

Training set: the dataset on which the algorithm is actually **trained**; it is used to calculate the gradient or Jacobian matrix and to **determine the update of the network weights in each iteration**.

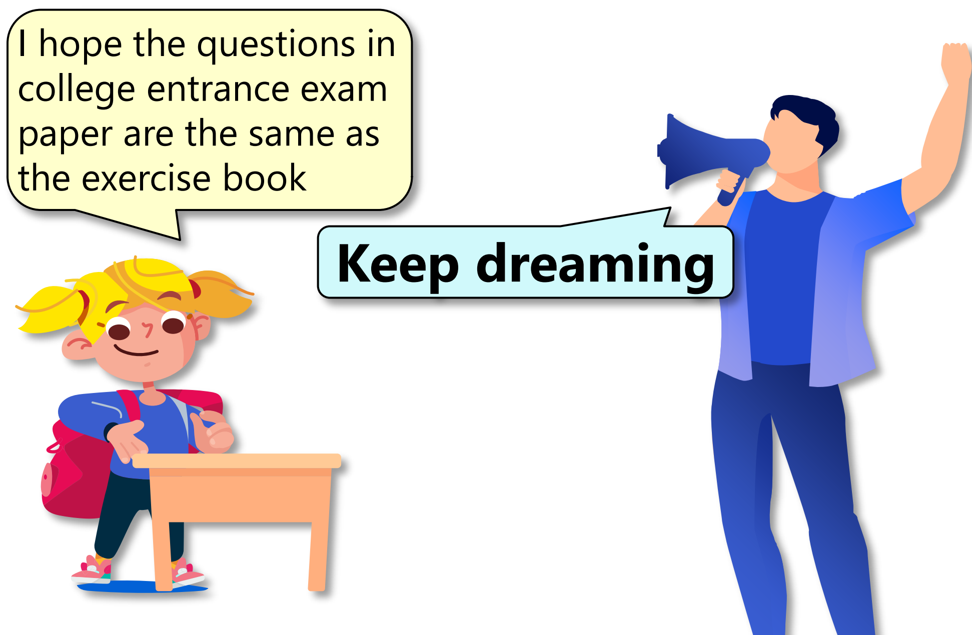
Validation set: the dataset used to **track its learning effect**; an indicator of what happened to the network function formed between the training data points, and the error values on the validation set will be monitored throughout the training process.

Test set: the dataset used to **generate the final results**.

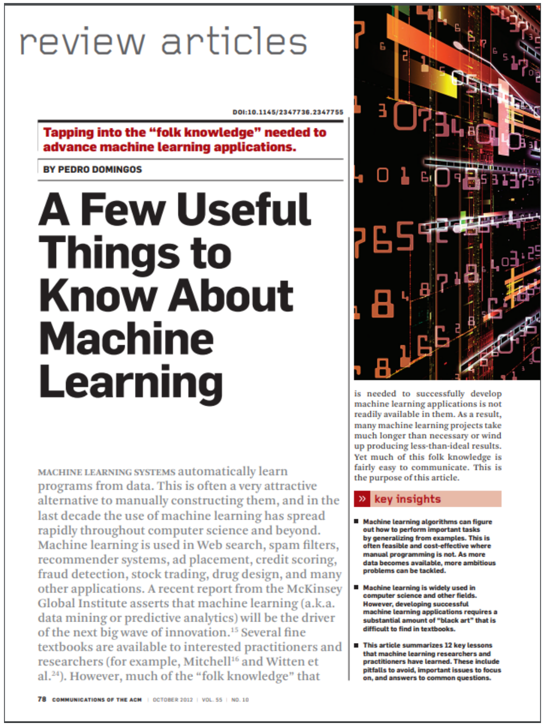
In order for the **test set to effectively reflect the generalization ability** of the network, it needs to be kept in mind:

First, the test set should never be used in any form to train a network, even if it is used to select a network from a set of alternative networks. Test sets should only be used after all training and model selection has been completed.

Second, the test set must be representative of all situations involved in the use of the network (difficult to guarantee when the input space is high-dimensional or complex in shape).



A Few Useful Things to Know About Machine Learning



review articles

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Tapping into the "folk knowledge" needed to advance machine learning applications.

BY PEDRO DOMINGOS

A Few Useful Things to Know About Machine Learning

MACHINE LEARNING SYSTEMS automatically learn programs from data. This is often a very attractive alternative to manually constructing them, and in the last decade the use of machine learning has spread rapidly throughout computer science and beyond. Machine learning is used in Web search, spam filters, recommender systems, ad placement, credit scoring, fraud detection, stock trading, drug design, and many other applications. A recent report from the McKinsey Global Institute asserts that machine learning (a.k.a. data mining or predictive analytics) will be the driver of the next big wave of innovation.[15] Several fine textbooks are available to interested practitioners and researchers (for example, Mitchell[16] and Witten et al.[24]). However, much of the "folk knowledge" that is needed to successfully develop machine learning applications is not readily available in them. As a result, many machine learning projects take much longer than necessary or wind up producing less-than-ideal results. Yet much of this folk knowledge is fairly easy to communicate. This is the purpose of this article.

» key insights

- Machine learning algorithms can figure out how to perform important tasks by generalizing from examples. This is often feasible and cost-effective where manual programming is not. As more data becomes available, more ambitious problems can be tackled.
- Machine learning is widely used in computer science and other fields. However, developing successful machine learning applications requires a substantial amount of "black art" that is difficult to find in textbooks.
- This article summarizes 12 key lessons that machine learning researchers and practitioners have learned. These include pitfalls to avoid, important issues to focus on, and answers to common questions.

Learning = Representation + Evaluation + Optimization

Suppose you have an application that you think machine learning might be good for. The first problem facing you is the **bewildering variety of learning algorithms available**. **Which one to use?** The key to not getting lost in this huge space is to realize that it consists of combinations of just **three components**:

Representation

A classifier must be represented in some formal language that the computer can handle. Conversely, choosing a representation for a learner is tantamount to choosing the set of classifiers that it can possibly learn. This set is called the **hypothesis space**. If a classifier is not in the hypothesis space, it cannot be learned. A related question is how to represent the input, what features to use.

Evaluation

An evaluation function (also called objective function or scoring function) is needed to distinguish good classifiers from bad ones. The evaluation function used internally by the algorithm may differ from the external one that we want the classifier to optimize, for ease of optimization and due to the issues.

Optimization

Finally, we need a method to search among the classifiers for the highest-scoring one. The choice of optimization technique is key to the efficiency of the learner, and also helps determine the classifier produced if the evaluation function has more than one optimum. It is common for new learners to start out using off-the-shelf optimizers, which are later replaced by custom-designed ones.

Table 1. The three components of learning algorithms.

Representation	Evaluation	Optimization
Instances	Accuracy/Error rate	Combinatorial optimization
K-nearest neighbor	Precision and recall	Greedy search
Support vector machines	Squared error	Beam search
Hyperplanes	Likelihood	Branch-and-bound
Naive Bayes	Posterior probability	Continuous optimization
Logistic regression	Information gain	Unconstrained
Decision trees	K-L divergence	Gradient descent
Sets of rules	Cost/Utility	Conjugate gradient
Propositional rules	Margin	Quasi-Newton methods
Logic programs		Constrained
Neural networks		Linear programming
Graphical models		Quadratic programming
Bayesian networks		
Conditional random fields		

PS A Few Useful Things to Know About Machine Learning

Learning = Representation + Evaluation + Optimization

Algorithm/Model	Representation	Evaluation	Optimization
Linear Regression	The model is defined as a linear equation, representing the target value via a linear combination of features.	Typically uses **Mean Squared Error (MSE)** to measure the difference between predicted and true values.	Commonly uses **gradient descent**, computing the gradient and iteratively updating parameters to minimize MSE.
Logistic Regression	Maps the probability value output by a linear function to the (0,1) interval, representing the probability that a sample belongs to the positive class.	Uses the **cross-entropy loss function (LogLoss)** to measure the discrepancy between the predicted probability distribution and the true label.	Commonly uses **gradient descent or its variants** (e.g., Newton's method) to minimize the cross-entropy loss and find the optimal parameters.
Decision Tree	Represents classification/regression rules via a tree structure. Internal nodes are tests on features, branches are outcomes of the tests, and leaf nodes are decisions.	Uses metrics such as **information gain** or **Gini impurity** to evaluate the purity of a split.	Employs **greedy search** (e.g., ID3, CART algorithms) to recursively select the best feature for splitting and build the tree.
Support Vector Machine	Defines a decision hyperplane in the feature space. Kernel tricks can represent complex non-linear boundaries.	Maximizes the **margin**, i.e., the minimum distance from any sample point to the hyperplane.	Solves a **convex quadratic programming problem** to find the optimal hyperplane that maximizes the margin.
Neural Network	A multi-layer network of neurons, performing complex non-linear function approximation through hierarchical layers.	Commonly uses **cross-entropy loss** or **mean squared error** to evaluate the difference between predictions and true labels.	Uses the **backpropagation algorithm** (often combined with stochastic gradient descent or its variants) to propagate errors backward and update weights.

PS Checking accuracy, completeness and F1

For the binary classification problem, the sample cases can be classified into **True Positive (TP), False Negative(FN), False Positive (FP), and True Negative(TN)** according to combining their **real categories** and the **predicted categories of the learner**, thus obtaining the following **Confusion matrix**

Real categories	Predicted categories	
	Positive Prediction	**Negative** Prediction
Positive Class	**TP**	**FN**
Negative Class	**FP**	**TN**

- **TP (True Positive)**: Shows the number of positive samples that are predicted to be positive. That is, if the true result is 1, the predicted result is also 1.
- **TN (True Negative):** Shows the number of negative samples predicted as negative cases. That is, if the true result is 0, the predicted result is also 0.
- **FP (False Positive):** Shows the number of negative samples that are predicted to be positive. That is, the true result is 0 and the predicted result is 1.
- **FN (False Negative):** Shows the number of positive samples that are predicted to be negative. That is, the true result is 1 and the predicted result is 0.

Accuracy: $acc = \frac{\mathrm{TP+TN}}{\mathrm{TP+TN+FP+FN}}$

Error rate: $err = \frac{\mathrm{FP+FN}}{\mathrm{TP+TN+FP+FN}}$

PS Checking accuracy, completeness and F1

Real categories	Predicted categories	
	Positive Prediction	**Negative** Prediction
Positive Class	**TP**	**FN**
Negative Class	**FP**	**TN**

Accuracy: $acc = \frac{TP+TN}{TP+TN+FP+FN}$

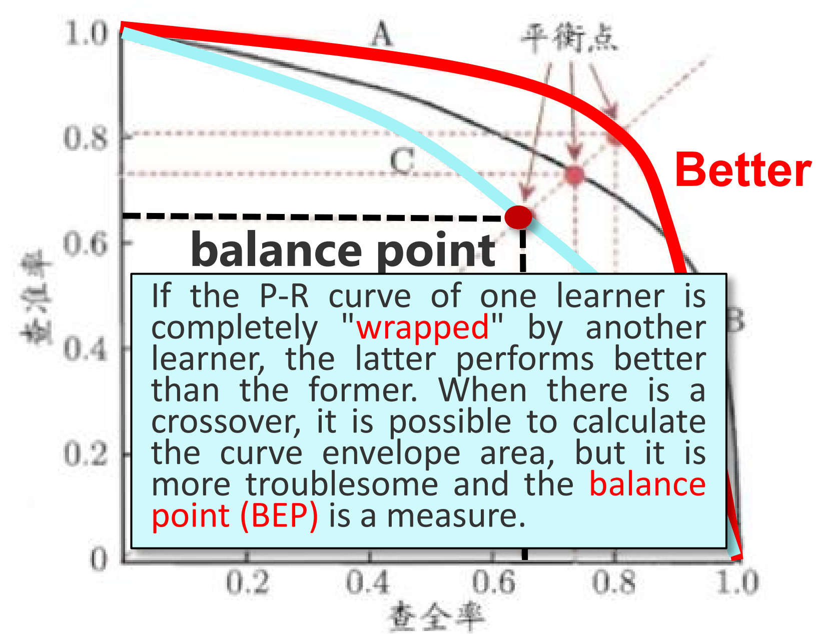
Error rate: $err = \frac{FP+FN}{TP+TN+FP+FN}$

Precision: abbreviated as P or PPV, indicates how many of the samples **predicted** to be positive (TP + FP) are **true positive** samples (TP).

P 查准率-预测出正例的正确率
宁缺毋滥

$$P = \frac{TP}{TP + FP}$$

The accuracy shows that it is better to miss the selection than wrong kill. The preferred way of thinking in identifying spam, because we do not want normal emails to be killed by mistake, which would cause serious problems.



Recall: abbreviated as R or TPR, indicates the percentage of the actual **true** positive sample (TP + FN) that is **predicted** to be **positive** (TP).

$$R = \frac{TP}{TP + FN}$$

The check-all rate indicates that it is better to make mistakes than to miss them. In the field of metal risk control, we hope that the system can screen out all risky behaviors or users, and then hand them over to the ministry to identify them, so that not to miss one may cause catastrophic result.

R 查全率-预测出正例的保证性
宁错杀一千，不放过1个

PS Checking accuracy, completeness and F1

However, the BEP is still a bit too simplified, and more commonly used are the F_1 and $F\beta$ measures, which are the **harmonic mean** and **weighted harmonic mean** of the Precision and Recall.

F_1

F_1 value is the harmonic mean of " precision and recall".

$$\frac{1}{F_1} = \frac{1}{2} \cdot \left(\frac{1}{P} + \frac{1}{R}\right)$$

$$F_1 = \frac{2 \times P \times R}{P + R} = \frac{2 \times TP}{number\ of\ samples + TP - TN}$$

When the F1 value of learner A is higher than that of learner B, then the BEP value of A is also higher than that of B (just substitute P=R into the F1 formula) (✓ or ×)

However, the F_1 value will give equal weight to both the the precision and recall. In some cases, you may want to favor one over the other, so we obtain a more normal **$F_\beta$value**

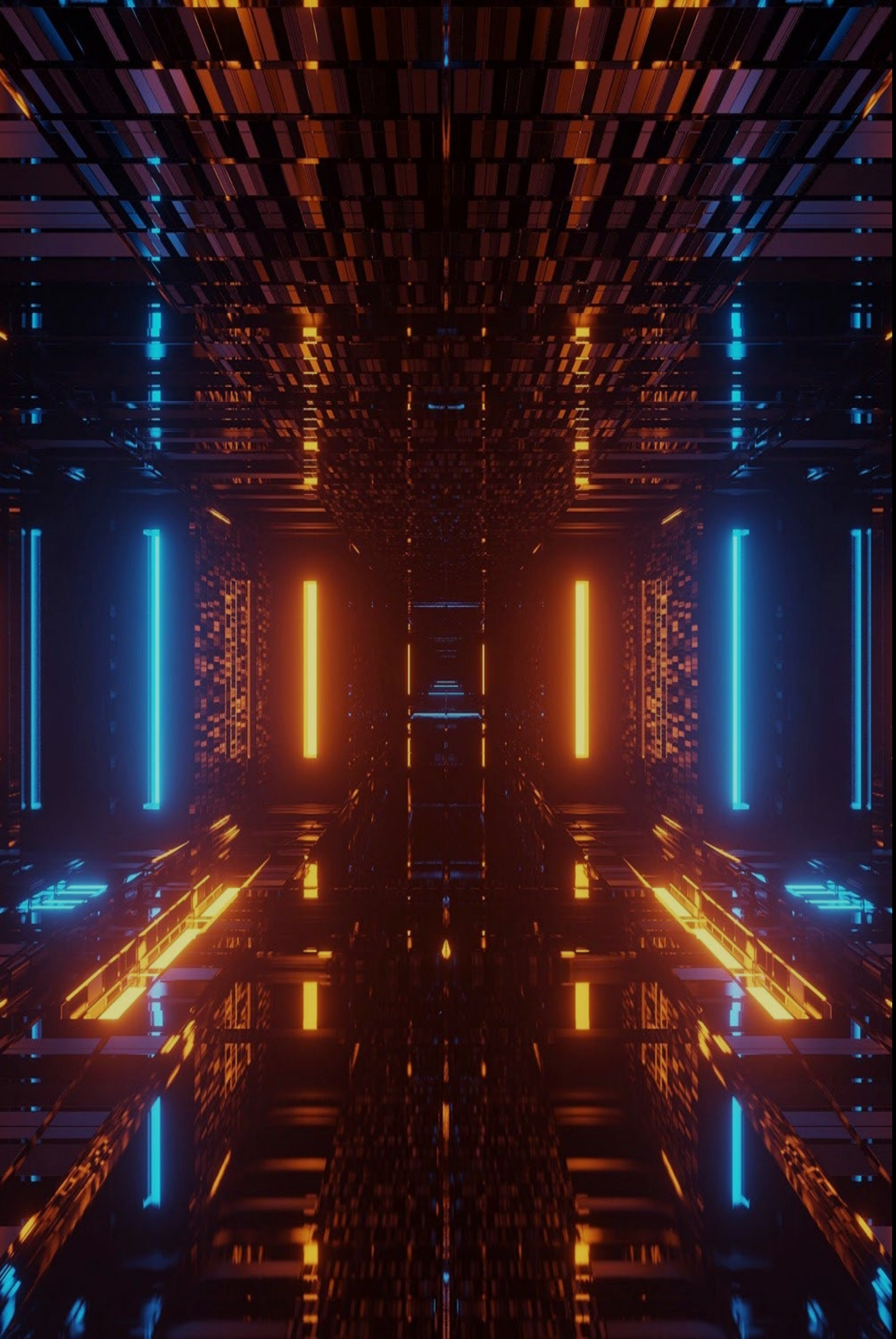
F_β

F_β value is the weighted harmonic mean of " precision and recall".

$$\frac{1}{F_\beta} = \frac{1}{1+\beta^2} \cdot \left(\frac{1}{P} + \frac{\beta^2}{R}\right)$$

$$F_\beta = \frac{(1+\beta^2) \times P \times R}{(\beta^2 \times P) + R}$$

Where β is used to adjust the weight, and when β = 1, both weights are the same, which is the F_1 value. If the **Precision rate is considered more important, β is reduced**; if the **Recall rate is considered more important, β is increased**.



Thanks